

# EXPLORATORY DATA ANALYSIS (EDA)

Dataset- Wine quality dataset

Source-

<https://archive.ics.uci.edu/ml/datasets/wine+quality>



# DESCRIPTION OF DATASET ATTRIBUTES

There are 11 columns describing their chemical properties, and a column for quality ratings. The dataset has following attributes:

- 1 - fixed acidity: most acids involved with wine or fixed or nonvolatile (do not evaporate readily)
- 2 - volatile acidity: the amount of acetic acid in wine, which at too high of levels can lead to an unpleasant, vinegar taste
- 3 - citric acid: found in small quantities, citric acid can add 'freshness' and flavor to wines
- 4 - residual sugar: the amount of sugar remaining after fermentation stops, it's rare to find wines with less than 1 gram/liter and wines with greater than 45 grams/liter are considered sweet
- 5 - chlorides: the amount of salt in the wine

## DESCRIPTION OF DATASET ATTRIBUTES

The dataset has following attributes:

- 6 - free sulfur dioxide: the free form of  $\text{SO}_2$  exists in equilibrium between molecular  $\text{SO}_2$  (as a dissolved gas) and bisulfite ion; it prevents microbial growth and the oxidation of wine
- 7 - total sulfur dioxide: amount of free and bound forms of  $\text{SO}_2$ ; in low concentrations,  $\text{SO}_2$  is mostly undetectable in wine, but at free  $\text{SO}_2$  concentrations over 50 ppm,  $\text{SO}_2$  becomes evident in the nose and taste of wine
- 8 - density: the density of water is close to that of water depending on the percent alcohol and sugar content

## **DESCRIPTION OF DATASET ATTRIBUTES**

The dataset has following attributes:

9 - pH: describes how acidic or basic a wine is on a scale from 0 (very acidic) to 14 (very basic); most wines are between 3-4 on the pH scale

10 - sulfates: a wine additive which can contribute to sulfur dioxide gas (SO<sub>2</sub>) levels, which acts as an antimicrobial and antioxidant

11 - alcohol: the percent alcohol content of the wine

Output variable (based on sensory data):

12 - quality (score between 0 and 10)

main goal is to discover which chemical properties influence the quality of red wines and to understand how these characteristics influence the quality.

Let's begin the code:

```
#import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from rpy2.robjects.lib.ggplot2 import ggplot
import seaborn as sns

#load the dataset
wine= pd.read_csv('Path_to_your_dataset/winequalityN.csv')
wine.head() #displaying top 5 entries of the dataset
```

|   | type  | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density | pH   | sulphates | alcohol | quality |
|---|-------|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|---------|------|-----------|---------|---------|
| 0 | white | 7.0           | 0.27             | 0.36        | 20.7           | 0.045     | 45.0                | 170.0                | 1.0010  | 3.00 | 0.45      | 8.8     | 6       |
| 1 | white | 6.3           | 0.30             | 0.34        | 1.6            | 0.049     | 14.0                | 132.0                | 0.9940  | 3.30 | 0.49      | 9.5     | 6       |
| 2 | white | 8.1           | 0.28             | 0.40        | 6.9            | 0.050     | 30.0                | 97.0                 | 0.9951  | 3.26 | 0.44      | 10.1    | 6       |
| 3 | white | 7.2           | 0.23             | 0.32        | 8.5            | 0.058     | 47.0                | 186.0                | 0.9956  | 3.19 | 0.40      | 9.9     | 6       |
| 4 | white | 7.2           | 0.23             | 0.32        | 8.5            | 0.058     | 47.0                | 186.0                | 0.9956  | 3.19 | 0.40      | 9.9     | 6       |

```
wine_red= wine[wine['type'].str.contains('red')] #filtering out the dataset of only red wine
len(wine_red) #length of the dataset
```

1599

```
#to check the number of entries of each quality
wine_red['quality'].value_counts()
```

```
5    681
6    638
7    199
4     53
8     18
3     10
Name: quality, dtype: int64
```

#find the correlation between data attributes for further exploration  
 wine\_red.corr()

|                      | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density   | pH        | sulphates | alcohol   | quality   |
|----------------------|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|-----------|-----------|-----------|-----------|-----------|
| fixed acidity        | 1.000000      | -0.255729        | 0.671184    | 0.114873       | 0.093358  | -0.154571           | -0.114297            | 0.667974  | -0.682458 | 0.184058  | -0.061390 | 0.123834  |
| volatile acidity     | -0.255729     | 1.000000         | -0.553022   | 0.001931       | 0.061503  | -0.011317           | 0.075770             | 0.022682  | 0.234546  | -0.261485 | -0.203671 | -0.390858 |
| citric acid          | 0.671184      | -0.553022        | 1.000000    | 0.143397       | 0.203792  | -0.059179           | 0.035781             | 0.364681  | -0.541346 | 0.314578  | 0.110798  | 0.226917  |
| residual sugar       | 0.114873      | 0.001931         | 0.143397    | 1.000000       | 0.055610  | 0.187049            | 0.203028             | 0.355283  | -0.085873 | 0.005577  | 0.042075  | 0.013732  |
| chlorides            | 0.093358      | 0.061503         | 0.203792    | 0.055610       | 1.000000  | 0.005562            | 0.047400             | 0.200632  | -0.264991 | 0.371574  | -0.221141 | -0.128907 |
| free sulfur dioxide  | -0.154571     | -0.011317        | -0.059179   | 0.187049       | 0.005562  | 1.000000            | 0.667666             | -0.021946 | 0.071091  | 0.050918  | -0.069408 | -0.050656 |
| total sulfur dioxide | -0.114297     | 0.075770         | 0.035781    | 0.203028       | 0.047400  | 0.667666            | 1.000000             | 0.071269  | -0.065558 | 0.042924  | -0.205654 | -0.185100 |
| density              | 0.667974      | 0.022682         | 0.364681    | 0.355283       | 0.200632  | -0.021946           | 0.071269             | 1.000000  | -0.341481 | 0.148881  | -0.496180 | -0.174919 |
| pH                   | -0.682458     | 0.234546         | -0.541346   | -0.085873      | -0.264991 | 0.071091            | -0.065558            | -0.341481 | 1.000000  | -0.197787 | 0.205818  | -0.057094 |
| sulphates            | 0.184058      | -0.261485        | 0.314578    | 0.005577       | 0.371574  | 0.050918            | 0.042924             | 0.148881  | -0.197787 | 1.000000  | 0.093331  | 0.251685  |
| alcohol              | -0.061390     | -0.203671        | 0.110798    | 0.042075       | -0.221141 | -0.069408           | -0.205654            | -0.496180 | 0.205818  | 0.093331  | 1.000000  | 0.476166  |
| quality              | 0.123834      | -0.390858        | 0.226917    | 0.013732       | -0.128907 | -0.050656           | -0.185100            | -0.174919 | -0.057094 | 0.251685  | 0.476166  | 1.000000  |



#let's see this correlation table in a more interesting way through a heat map

k = 12

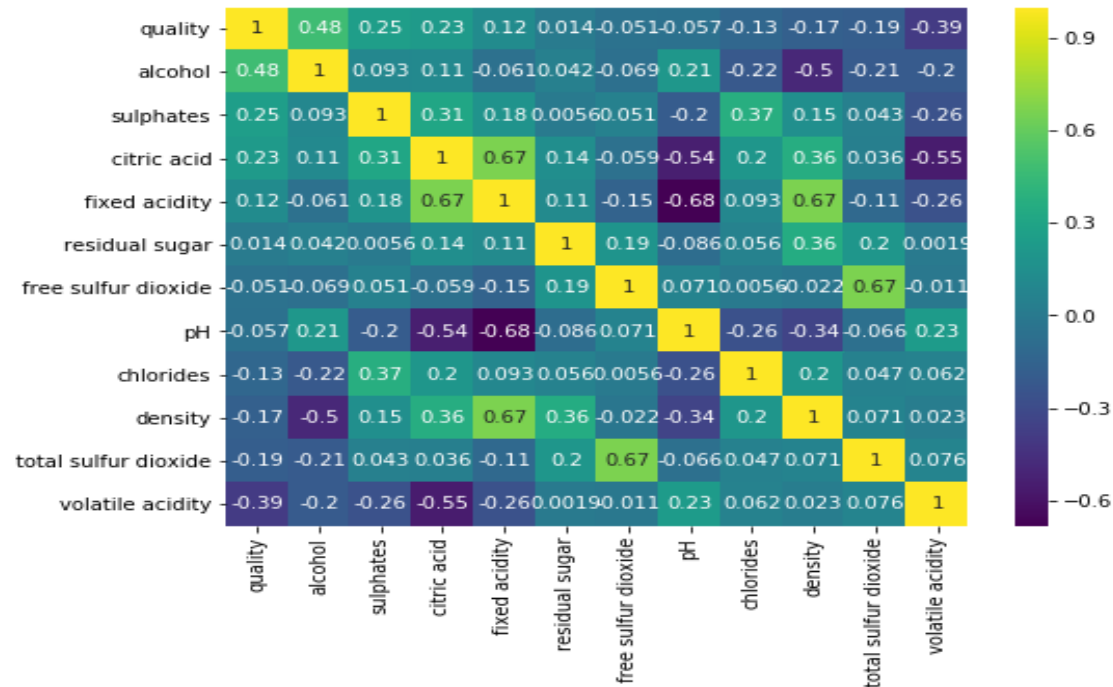
cols = wine\_red.corr().nlargest(k, 'quality')['quality'].index

cm = wine\_red[cols].corr()

plt.figure(figsize=(8,6))

sns.heatmap(cm, annot=True, cmap = 'viridis')

<matplotlib.axes.\_subplots.AxesSubplot at 0x29e8aa5bcc0>



## Conclusion from the correlation graph

As it can be seen from the graph this is not a very effective correlation between any of the attributes or between an attribute and quality of the wine.

The following dimensions are relatively highly correlated:

total.sulfur.dioxide with free.sulfur.dioxide; fixed.acidity with density and citric.acid

The following dimensions are relatively correlated:

alcohol with quality (this might be a candidate for drop, since might be a leak)

The following dimensions are relatively highly inverse correlated:

fixed.acidity with pH

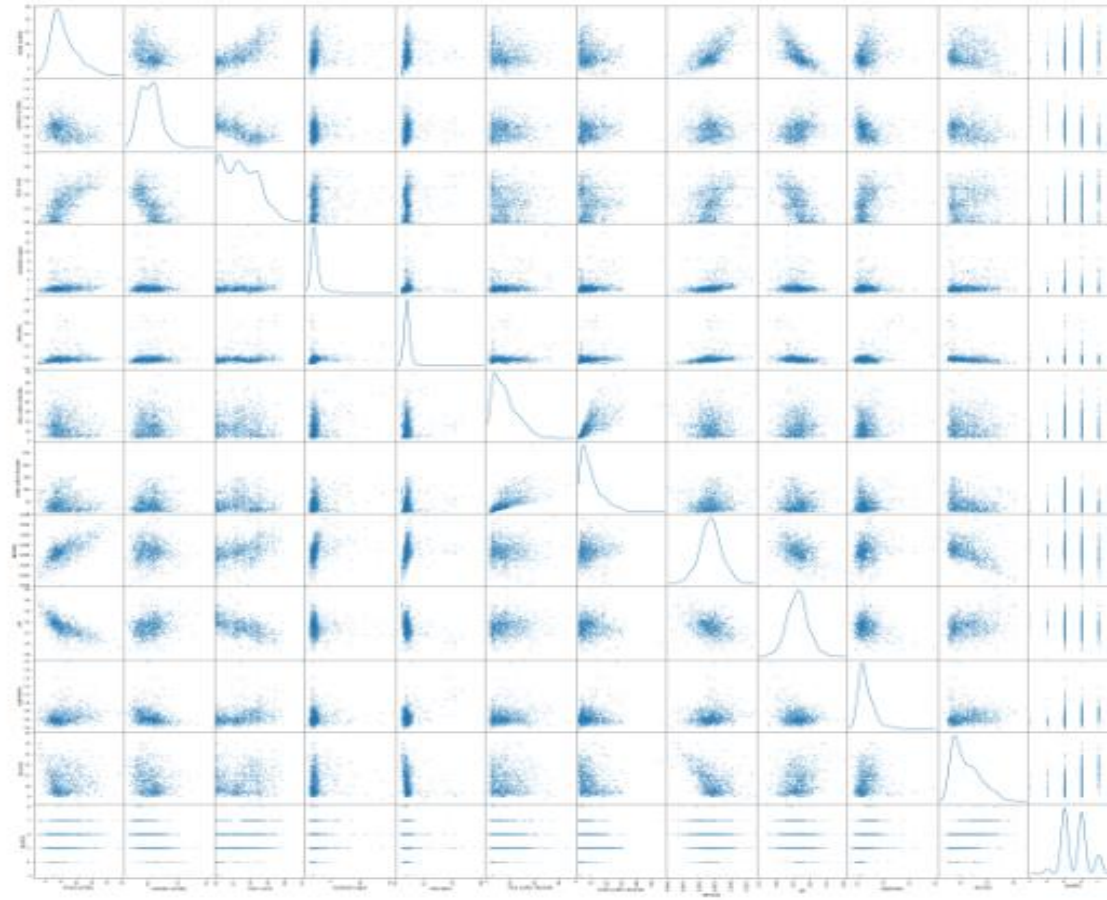
The following dimensions are relatively inverse correlated:

citric.acid with pH and volatile.acidity

As we see, the top 3 come to be: alcohol, volatile acidity, citric acid

#plotting scatter matrix to know about the distribution of each attribute

```
pd.plotting.scatter_matrix(wine_red, alpha = 0.3, figsize = (40,40), diagonal = 'kde')
```

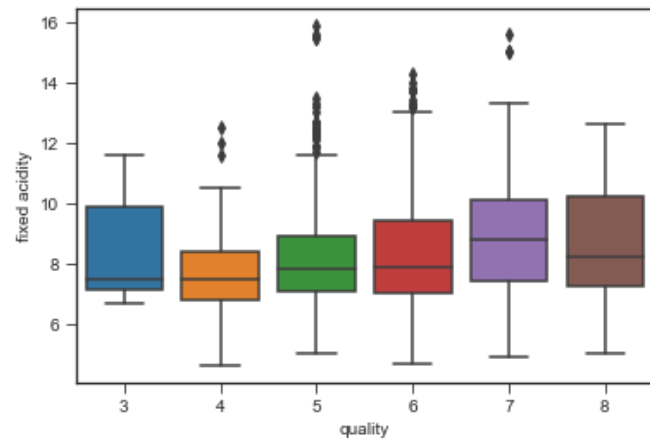


## Conclusion from the plot

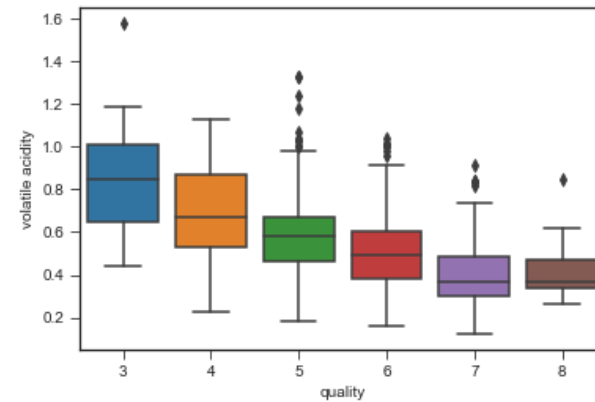
- It appears that density and pH are normally distributed, with a few outliers.
- Fixed and volatile acidity, sulfur dioxides, sulphates, and alcohol seem to be long-tailed.
- Qualitatively, residual sugar and chlorides have extreme outliers.
- Citric acid appeared to have a large number of zero values. This might be a case of non reporting.

So now lets check the relation of each variable with the variable wine quality to test the outliers in the data

```
sns.boxplot('quality','fixed acidity', data=wine_red)
```

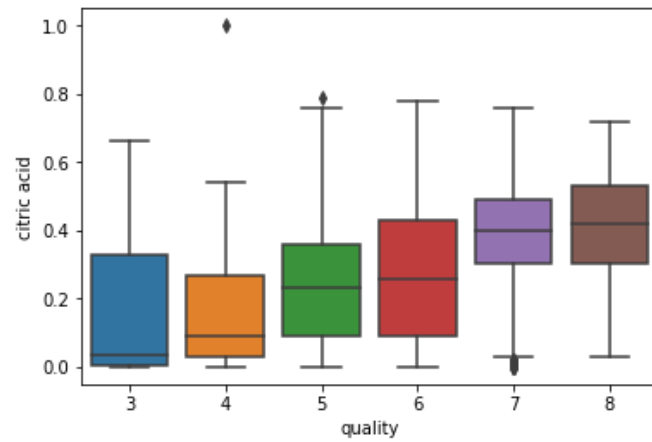


```
sns.boxplot('quality','volatile acidity', data=wine_red)
```

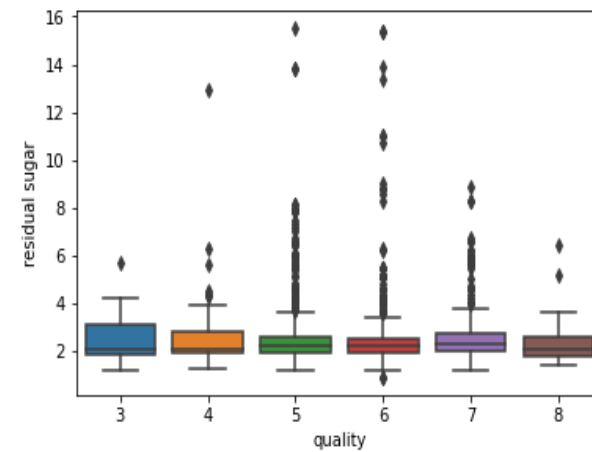


# Boxplot

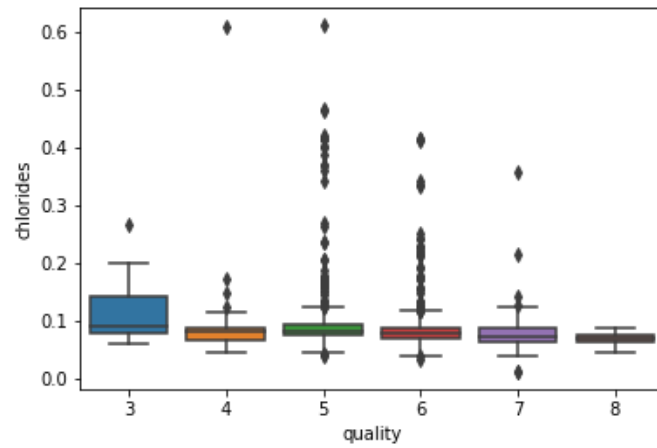
```
sns.boxplot('quality','citric acid', data=wine_red)
```



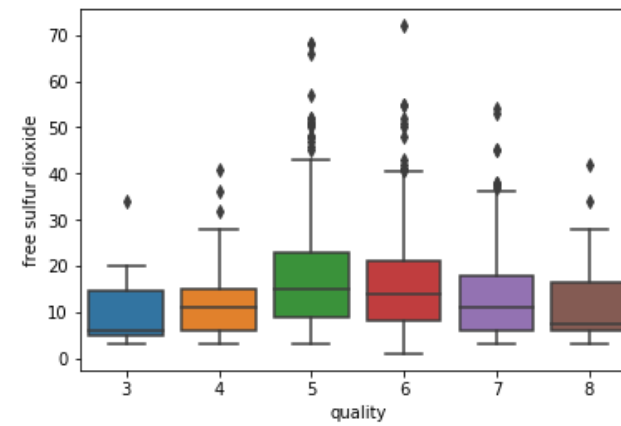
```
sns.boxplot('quality','residual sugar', data=wine_red)
```



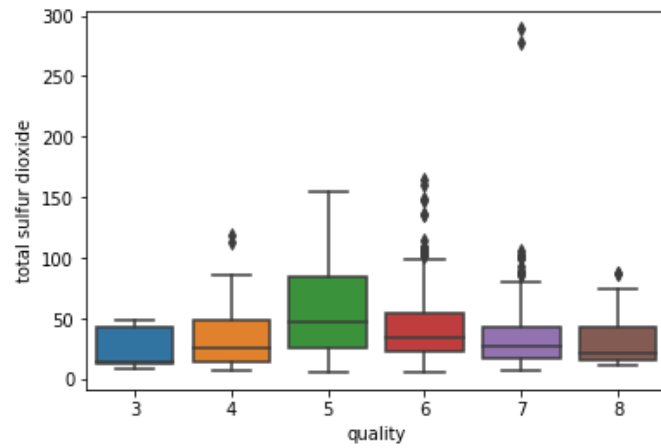
```
sns.boxplot('quality','chlorides', data=wine_red)
```



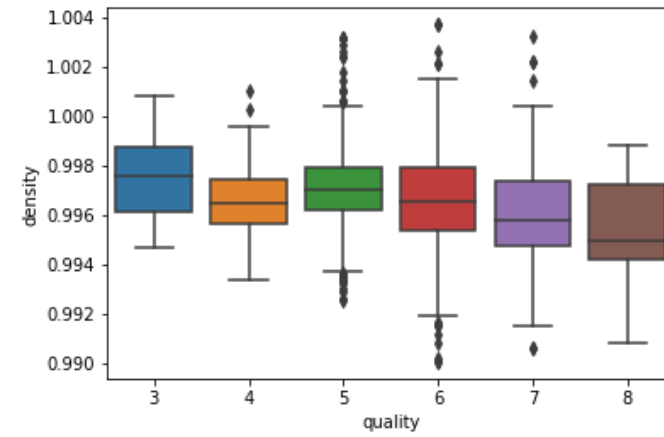
```
sns.boxplot('quality','free sulfur dioxide', data=wine_red)
```



```
sns.boxplot('quality','total sulfur dioxide', data=wine_red)
```

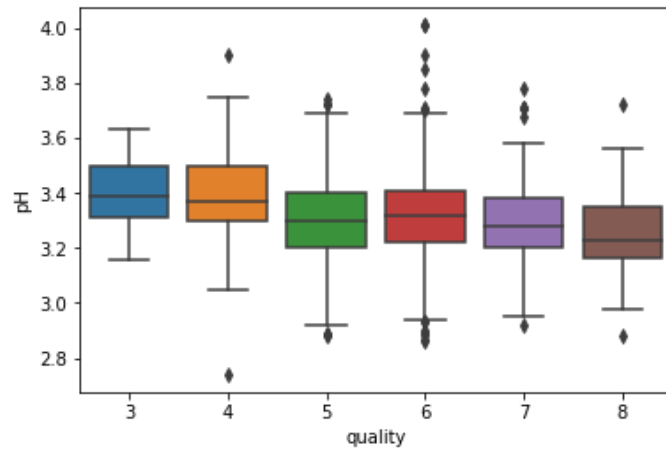


```
sns.boxplot('quality','density', data=wine_red)
```

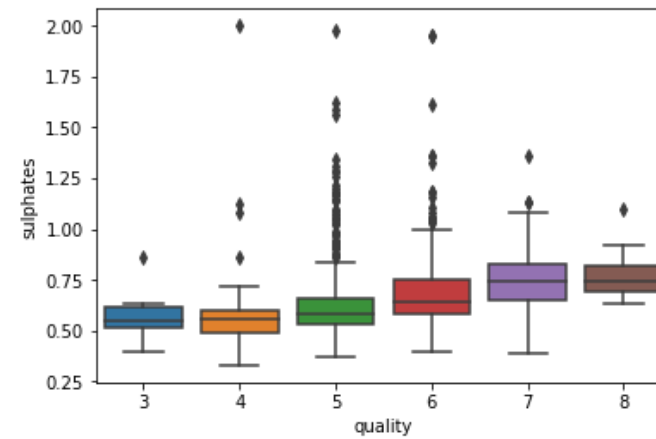




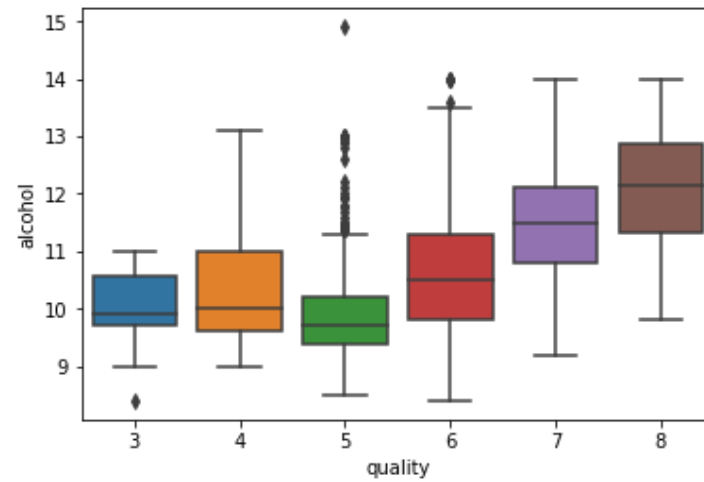
```
sns.boxplot('quality','pH', data=wine_red)
```



```
sns.boxplot('quality','sulphates', data=wine_red)
```



```
sns.boxplot('quality','alcohol', data=wine_red)
```



boxplots show many outliers for quite a few columns. Describe the dataset to get a better idea on what's happening

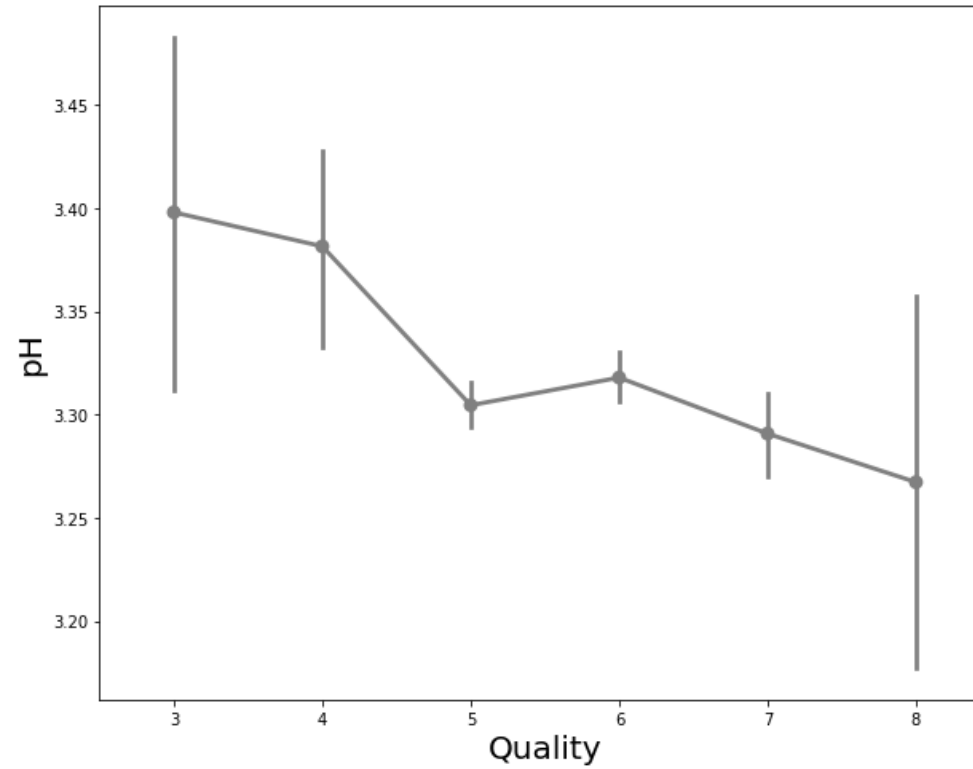
wine\_red.describe()

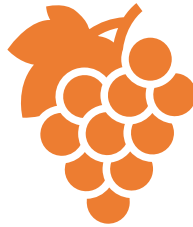
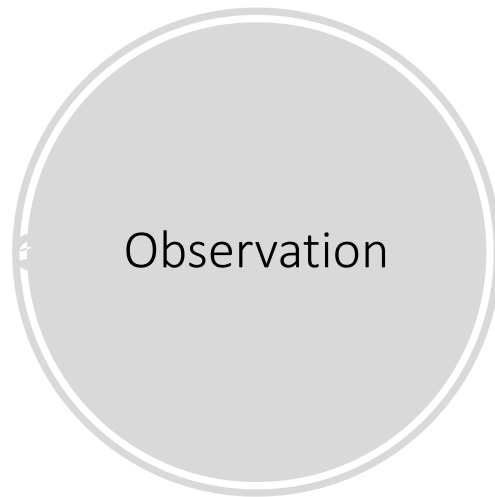
|       | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides   | free sulfur dioxide | total sulfur dioxide | density     | pH          | sulphates   | alcohol     |      |
|-------|---------------|------------------|-------------|----------------|-------------|---------------------|----------------------|-------------|-------------|-------------|-------------|------|
| count | 1597.000000   | 1598.000000      | 1598.000000 | 1599.000000    | 1599.000000 | 1599.000000         | 1599.000000          | 1599.000000 | 1597.000000 | 1597.000000 | 1599.000000 | 1599 |
| mean  | 8.322104      | 0.527738         | 0.271145    | 2.538806       | 0.087467    | 15.874922           | 46.467792            | 0.996747    | 3.310864    | 0.658078    | 10.422983   | 5    |
| std   | 1.740767      | 0.179085         | 0.194744    | 1.409928       | 0.047065    | 10.460157           | 32.895324            | 0.001887    | 0.154290    | 0.169594    | 1.065668    | 0    |
| min   | 4.600000      | 0.120000         | 0.000000    | 0.900000       | 0.012000    | 1.000000            | 6.000000             | 0.990070    | 2.740000    | 0.330000    | 8.400000    | 3    |
| 25%   | 7.100000      | 0.390000         | 0.090000    | 1.900000       | 0.070000    | 7.000000            | 22.000000            | 0.995600    | 3.210000    | 0.550000    | 9.500000    | 5    |
| 50%   | 7.900000      | 0.520000         | 0.260000    | 2.200000       | 0.079000    | 14.000000           | 38.000000            | 0.996750    | 3.310000    | 0.620000    | 10.200000   | 6    |
| 75%   | 9.200000      | 0.640000         | 0.420000    | 2.600000       | 0.090000    | 21.000000           | 62.000000            | 0.997835    | 3.400000    | 0.730000    | 11.100000   | 6    |
| max   | 15.900000     | 1.580000         | 1.000000    | 15.500000      | 0.611000    | 72.000000           | 289.000000           | 1.003690    | 4.010000    | 2.000000    | 14.900000   | 8    |

- Fixed acidity - 25% - 7.1 and 50% - 7.9. Not much of a variance. Could explain the huge number of outliers volatile acidity
- Similar reasoning citric acid - seems to be somewhat uniformly distributed residual sugar - min - 0.9, max - 15!! Way too much difference. Could explain the outliers.
- Chlorides - same as residual sugar. Min - 0.012, max - 0.611
- Free sulfur dioxide, total sulfur dioxide - same explanation as above

Let's check a  
relationship between  
pH/acidity and quality  
of wine

```
plt.figure(figsize=(10,8))  
sns.pointplot(wine_red['quality'],  
wine_red['pH'], color='grey')  
plt.xlabel('Quality').set_size(20)  
plt.ylabel('pH').set_size(20)
```





Great wines are in balance with their 4 fundamental traits (Acidity, tannin, alcohol and sweetness) and as wines age, the acidity acts as a buffer to preserve the wine longer.



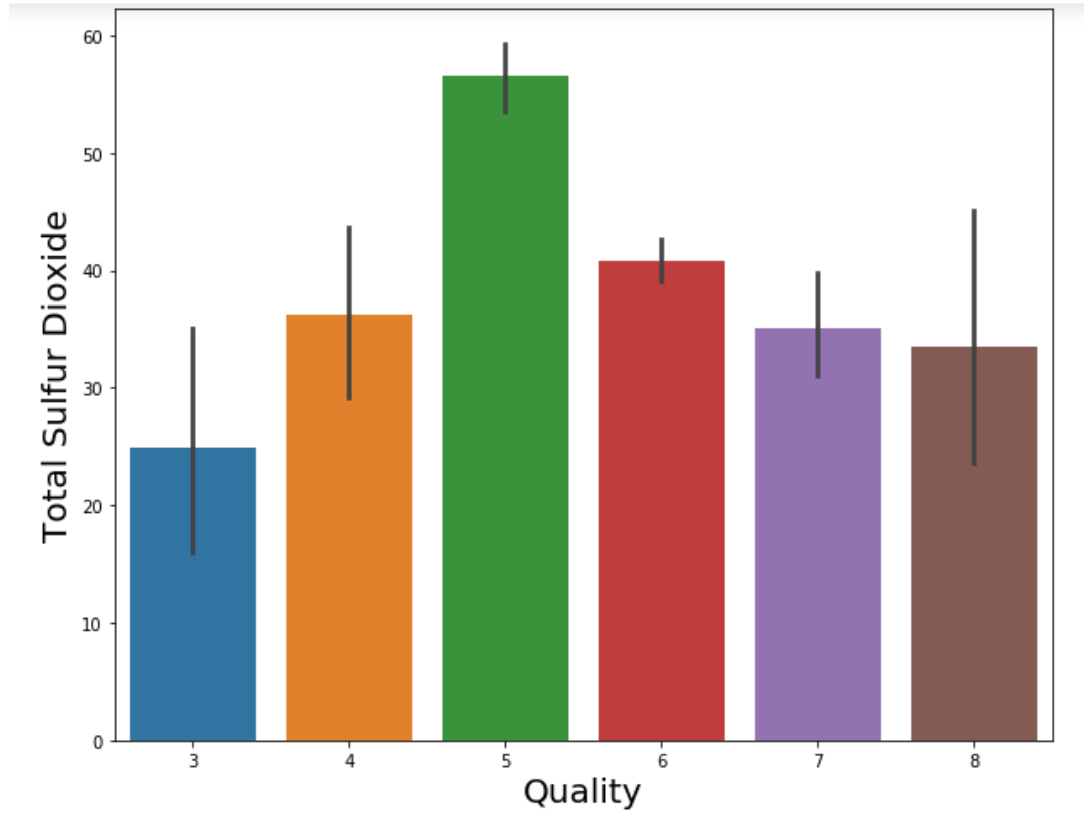
We can infer from the graph that higher quality wines have a comparatively low pH/high Acidity

## Exploring some more relationships

Many would be aware that sulfur dioxide can cause severe allergic reactions in some people. So why do winemakers use it? Put simply, it is very difficult to make wines that have an aging potential beyond a few months if sulfur dioxide is not used during winemaking.

```
plt.figure(figsize=(10,8))  
sns.barplot(wine_red['quality'],wine_red['total sulfur dioxide'])  
plt.xlabel('Quality').set_size(20)  
plt.ylabel('Total Sulfur Dioxide').set_size(20)
```

Therefore, we can see from the graph that wines do include a good quantity of sulfur dioxide in them.



## Let's check relationship between wine pH and residual sugar

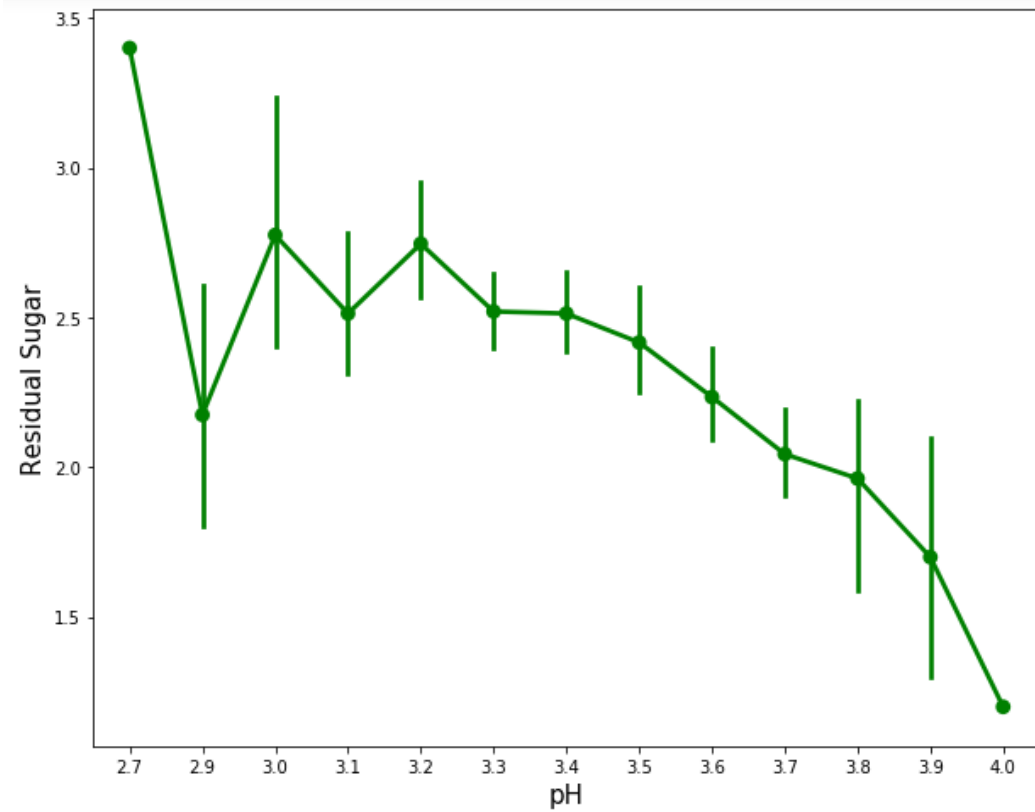
Sugar also has a bonus effect: it can help wines age well.

Whether your bottle is young or elderly, you should think of RS as having a balancing relationship with acidity. They are on opposite sides of the seesaw, so if the wine has sugar you will probably want strong acidity, too—otherwise the wine will feel cloying.

```
plt.figure(figsize=(10,8))
sns.pointplot(wine['pH'].round(1),wine['residual sugar'], color='green')
plt.xlabel('pH').set_size(15)
plt.ylabel('Residual Sugar').set_size(15)
```



From the graph, we can see that low levels of pH(High Acidity) have high levels of Sugar.



## Reflections

- In this analysis, we firstly explore all attributes by their distributions and frame the questions we are interested in. For example, can sweetness, pH and citric acid improve quality?
- Then, we created correlation matrix, select the most 3 significant attributes (alcohol, sulphates, volatile.acidity) associated with quality, and then explore how the data distributes along these 3 attribute dimensions.
- Next, we plotted how these 3 attributes associate with quality by scatter plot and linear model lines: - in alcohol vs. quality plot and sulphates vs. quality plot, we found both have positive association, though the association is not very strong - in volatile.acidity, the association is negative.
- Furthermore, we also use leveled scatter plot to get idea how quality level distributed in different attributes' dimension scales and concluded that: Higher pH, higher sulphates, higher alcohol and higher fixed acidity are more likely to indicate red wines with higher quality.